

**STATEMENT OF WORK:**  
**“Quarters 6 Plumbing Replacement, Mold  
Remediation, HVAC Installation, and Crawlspace  
Encapsulation”**

**DATED:**  
**2026**



**Stones River National Battlefield**  
**3501 Old Nashville Hwy Murfreesboro, TN.37129**

## **1. Summary of Work**

This project consists of comprehensive rehabilitation of Quarters 6, a 1,750 SF, 3-bedroom, 1.5-bathroom residential structure. The project includes demolition of deteriorated plumbing and HVAC components, mold remediation, installation of new PVC plumbing (interior and exterior), HVAC duct system replacement, crawlspace encapsulation, and a full testing and commissioning. Location: 103 Old Lodge Lane, Murfreesboro, TN 37129.

The work shall include removal of cast iron sewer piping, correction of sagging main line, installation of cleanouts, installation of pressure-assisted toilets, installation of ductwork, crawlspace vapor barrier encapsulation (minimum 14-mil), rigid wall insulation, and installation of a dehumidification system.

All work shall comply with applicable International Residential Code (IRC), International Plumbing Code (IPC), International Mechanical Code (IMC), OSHA 29 CFR 1926, and all local and state building codes.

The contractor shall maintain continuous access and utility service to adjacent facilities throughout the duration of the project. Access to the Natural Resource Management (NRM) building (111 Old Lodge Lane), the Maintenance Yard (115 Old Lodge Lane), and Quarters 5 (Q5) (107 Old Lodge Lane) shall not be obstructed or restricted without prior coordination and approval from the NPS Facility Manager. Sewer services to the NRM building, Maintenance Yard, and Q5 shall remain operational at all times during construction. Any required temporary shutdowns shall be minimized during duration and scheduled in advance with written approval from the Contracting Officer's Representative (COR).

## **2. Project Tasks**

### **Demolition and Removal – Contractor Shall:**

- Remove all existing HVAC ductwork within the crawlspace.
- Remove approximately:
  - 36 LF of 4" cast iron pipe from interior.
  - 20 LF of cast iron pipe and 20 LF of PVC pipe from exit point to past the sag section.
- Remove existing:
  - 2 Toilets
  - 2 Flanges
  - 2 Traps
  - All associated plumbing fixtures
- Remove all debris from crawlspace.
- Level crawlspace floor and backfill any existing animal intrusion holes.
- Properly dispose of all demolition debris in approved landfill facilities.
- Protect structural components and utilities during demolition.
- Follow OSHA demolition safety standards (29 CFR 1926 Subpart T).

### **Mold Remediation – Contractor Shall:**

- Install negative air machines in the crawlspace during remediation activities.
- Apply approved microbial growth treatment to crawlspace surfaces.
- Remove negative air machines after remediation is complete.
- Dispose of contaminated materials in accordance with EPA mold remediation guidelines.

### **Interior PVC Plumbing Installation – Contractor Shall:**

- Install new PVC plumbing system servicing:
  - Two new Kohler Arch 12 in. 1.28 GFP single flush toilets or equivalent
  - Three existing sinks
- Connect 1.5 bathrooms, sinks, and kitchen sink to new PVC main exiting the home.
- Inspect subfloor at toilet flange locations; replace deteriorated subfloor as necessary.
- Repair any wall or floor surface that required to be removed related to access for the plumbing install.
- Install two new pressure-assisted toilets (one per bathroom).
- Ensure proper slope of horizontal drainage lines per IPC.
- Test all plumbing for leaks and proper drainage.

### **Exterior PVC Plumbing Installation – Contractor Shall:**

- Use a camera to identify and mark the length of the sag in the pipe.
- Excavate approximately:
  - 20 LF of cast iron pipe
  - 20 LF of PVC pipe
- Use concrete saw to remove two sidewalk sections as required.
- Replace cast iron with PVC pipe
- Correct sag (belly) to ensure proper gravity flow in accordance with IPC.
- Install:
  - One cleanout near house
  - Once cleanout near belly section area
- Backfill trench per building code requirements.
- Repair the two sections of sidewalk that were removed.
- Compact trench backfill in lifts to prevent settlement.
- Patch removed concrete sections
- Seed and straw disturbed ground after final grading

### **HVAC Ductwork Installation– Contractor Shall:**

- Install complete ductwork system for 1,750 SF house (3BR, 1.5BA).
- Insulate all ductwork to prevent condensation.
- Secure ductwork with hangers at proper spans per IMC.
- Connect ductwork to existing 1-ton air-cooled electric AC unit.
- Seal all duct connections to prevent air leakage.
- Balance airflow system.

### **Crawlspace Encapsulation – Contractor Shall:**

- Seal all crawlspace vents and openings from interior.
- Seal foundation cracks using caulk or hydraulic cement if present.
- Install minimum 14-mil polyethylene vapor barrier on:
  - Crawlspace floor
  - Crawlspace walls
- Install rigid foam insulation on crawlspace walls.
- Install minimum 70 PPD Aprilaire dehumidifier (or equivalent) with:
  - Condensate pump discharge
  - Dedicated electrical outlet
  - Wall-mounted controller in utility closet
- Properly dispose of all construction debris.

### **Testing and Commissioning – Contractor Shall:**

- Test plumbing system for leaks and proper drainage.
- Test HVAC system for airflow and performance.
- Test dehumidifier operation and controller functionality.
- Provide written confirmation of system testing and performance.

### **3. Submittals**

- Contractor shall submit safety plan for approval.
- Plumbing layout drawing
- HVAC duct layout
- Dehumidifier product data
- Mold remediation product data
- Test reports for plumbing and HVAC.
- Product data for all next fixtures (e.g. toilets)
- All submittals shall be on the Government CM-16 form and approved by the project COR.

### **4. Security**

- The contractor is responsible for all materials and equipment on the job site at all times.
- All employees working on the job site shall be legal to work in the United States of America and not be a prior convicted felon.
- Park hours are from 7:00 AM until 4:00 PM. The Contractor may plan to work on site from 7 AM to 4:00 PM Monday thru Friday; except for Federal holidays or when the Government facility is closed due to local or national emergencies, administrative closings, or similar Government directed facility closings. Contractor shall not work outside of these hours, without prior approval from STRI Facility Manager.

### **5. Administrative Requirements**

- Contractor shall provide all safety equipment to include warning cones and barricades while on site to protect the public and staff members while work is ongoing.
- Jobsite superintendent shall ensure all jobsite personnel are following the site safety plan as submitted and approved.
- Contractor shall sweep the area of dirt and debris at the conclusion of the work.

### **6. Substantial Completion and Final Inspection**

- Contractor shall submit a written request that the project is complete and request a final inspection. The COR will complete the inspection and develop a punch list if there is work remaining
  - If the COR determines that the work is not complete, he will immediately notify the Contractor, in writing, stating the reasons. After completing the work, the Contractor shall resubmit a request for another final inspection.

## **7. Post-Construction Inspection**

The COR and the contractor's project manager shall conduct the walk around to do a final inspection after the contractor notifies the COR that the punch list is complete. Any, and all deficiencies shall be corrected before the National Park Service will accept the project.

## **8. Acceptance of the Work**

After all deficiencies have been corrected, and work has been accepted by the Government, payment will be made in accordance with the Schedule of Values.

## **9. Measure and Payment**

Payment will be included under the lump sum price for this project. The contractor must ensure that they are registered in the Invoice Processing Platform at [www.ipp.gov](http://www.ipp.gov) as that is the platform to submit invoices and receive payment.

## **10. Contracted Services Statement**

Contractor's employees, performing services under this award, shall be controlled, directed, and always supervised by the assigned project manager that is fluent in the English Language. Contractor's management staff shall ensure that employees properly comply with the performance work standards outlined in the scope of work, specifications, and contract documents. The tasks, duties, and responsibilities set forth in the Statement of Work, and Specifications may not be interpreted or implemented in any manner that results in any contractor employee creating or modifying Federal policy, obligating the appropriated funds of the United States Government, overseeing the work of Federal employees, providing direct personal services to any Federal employee, or otherwise violating the prohibitions set forth in Parts 7.5 and 37.1 of the Federal Acquisition Regulation (FAR).

# CM-16 TRANSMITTAL FORM - NATIONAL PARK SERVICE

|                         |                         |                                                          |                           |                                                                                                                                              |                                                                    |                                                                                                                                                                                                                |                                                   |                             |                              |                        |          |                                  |                        |                         |        |
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| Park:                   |                         | PMIS No.:                                                |                           | Project:                                                                                                                                     |                                                                    |                                                                                                                                                                                                                | Contract No.:                                     |                             |                              | Date:                  |          | Transmittal No.:<br>Sheet: 1 of: |                        |                         |        |
| Contractor:             |                         |                                                          |                           |                                                                                                                                              | Subcontractor / Supplier:                                          |                                                                                                                                                                                                                |                                                   |                             |                              | NPS ACTION             |          |                                  |                        |                         |        |
|                         |                         |                                                          |                           |                                                                                                                                              |                                                                    |                                                                                                                                                                                                                |                                                   |                             |                              | Action Submittal       |          |                                  |                        | Informational Submittal |        |
| Action Submittal        | Informational Submittal | Item No.                                                 | Specification Section No. | Paragraph No.                                                                                                                                | Description of Item<br>(Size, Type, Name, Manufacturer, Use, Etc.) | No. of Copies Submitted                                                                                                                                                                                        | Contributes to meeting LEED and/or Fed Sust. Req. | Meets Buy American Act Req. | Alternate Materials Proposed | No. of Copies Returned | Approved | Approved with Notations          | Disapproved - Resubmit | Accept                  | Reject |
|                         |                         |                                                          |                           |                                                                                                                                              |                                                                    |                                                                                                                                                                                                                |                                                   |                             |                              |                        |          |                                  |                        |                         |        |
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| Contractor's Signature: |                         |                                                          |                           |                                                                                                                                              |                                                                    | Recommended by:                                                                                                                                                                                                |                                                   |                             |                              |                        |          |                                  |                        |                         |        |
| Title:                  |                         | Date:                                                    |                           | I hereby certify that this submittal has been reviewed for accuracy, completeness, and compliance with contract requirements (FAR 52.236-21) |                                                                    |                                                                                                                                                                                                                |                                                   |                             |                              | Title:                 |          |                                  |                        | Date:                   |        |
| Review Comments:        |                         |                                                          |                           |                                                                                                                                              |                                                                    | Action by:                                                                                                                                                                                                     |                                                   |                             |                              |                        |          |                                  |                        |                         |        |
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|                         |                         |                                                          |                           |                                                                                                                                              |                                                                    | Approval of this submittal is subject to the provisions of the contract drawings and specifications. This action is for general concurrence only and the government is not responsible for errors or omissions |                                                   |                             |                              |                        |          |                                  |                        |                         |        |
| Date Received:          |                         | Distribution: 1 copy each: CMR Interim, A/E, DSC, Contr. |                           |                                                                                                                                              |                                                                    |                                                                                                                                                                                                                |                                                   |                             |                              |                        |          |                                  |                        |                         |        |
| Date To Contractor:     |                         |                                                          |                           |                                                                                                                                              |                                                                    |                                                                                                                                                                                                                |                                                   |                             |                              | r03/10                 |          |                                  |                        |                         |        |

# CM-16 TRANSMITTAL FORM CONTINUATION - NATIONAL PARK SERVICE

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|-------|-----------|----------|---------------|-------|-------------------|
| Park: | PMIS No.: | Project: | Contract No.: | Date: | Transmittal No.:  |
|       |           |          |               |       | Sheet:        of: |

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|-------------|---------------------------|------------------|-------------------------|
| Contractor: | Subcontractor / Supplier: | NPS ACTION       |                         |
|             |                           | Action Submittal | Informational Submittal |

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| Review Comments |
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